Department of Computer Applications

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Synopsis of Minor Project

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➢ Title of the Project

Restaurant Searching and Ranking System

1. Introduction

1.1 Purpose

The purpose of the Restaurant Searching and Ranking System is to help users find the best and most suitable restaurants according to their needs and preferences. Nowadays, people often face difficulties in choosing a restaurant because there are many restaurants available with different cuisines, prices, locations, ratings, and reviews. Searching for the right restaurant can take a lot of time.

This system provides a simple and convenient way to search for restaurants and compare their details in one place. Users can search for restaurants based on their location, preferred food type, budget, and ratings. The system displays a list of matching restaurants and ranks them according to selected criteria, helping users identify suitable options more easily.

The system aims to save users' time, reduce the difficulty of finding restaurants, and help them make better decisions. It also provides restaurant information in an organized manner so that users can compare different options before selecting a restaurant.

1.2 Scope

The Restaurant Searching and Ranking System is a web-based application that allows users to search for restaurants and compare their details.

The system provides the following features:

- Search for restaurants by name, location, or cuisine.
- View restaurant details, ratings, prices, and reviews.
- Rank restaurants based on ratings and user preferences.
- Filter restaurants according to price, location, and cuisine.
- Display the most suitable restaurants at the top of the results

1.3 Objectives

The main objectives of the system are:

1. To help users find suitable restaurants easily.
2. To provide accurate restaurant information.
3. To rank restaurants based on ratings and user preferences.
4. To save users' time when searching for restaurants.
5. To allow users to compare different restaurants.
6. To improve the restaurant selection process.

1.4 Intended Audience

The intended audience of this system includes:

- Customers: People who want to find suitable restaurants.
- Restaurant Owners: People who want to display their restaurant details.
- Administrators: People who manage restaurant information and system data.
- Developers: People who develop and maintain the system.

1.5 Document Overview

This SRS document describes the requirements of the Restaurant Searching and Ranking System.

It explains the system's purpose, scope, objectives, functional requirements, non-functional requirements, hardware and software requirements, business rules, future enhancements, and conclusion.

It helps developers understand how the system should work.

1.6 Foundational Concepts Used

The system uses the following basic concepts:

1. **Restaurant Searching:** Allows users to find restaurants using keywords such as name, location, and cuisine.
2. **Ranking System:** Arranges restaurants according to ratings, reviews, price, and user preferences.
3. **Filtering:** Allows users to narrow search results based on specific requirements.
4. **Database:** Stores restaurant details, ratings, reviews, and other information.
5. **User Interface:** Provides an easy-to-use interface for searching and viewing restaurants.
6. **Recommendation System:** Suggests suitable restaurants based on user preferences.

2. Specific Requirements

2.1 Functional Requirements

Functional requirements describe **what the system should do**.

The main functional requirements are:

1. **User Search:** The system should allow users to search for restaurants by name, location, or cuisine.
2. **Restaurant Details:** The system should display restaurant names, addresses, cuisine types, prices, and ratings.
3. **Restaurant Ranking:** The system should rank restaurants based on ratings and selected preferences.
4. **Filtering:** The system should allow users to filter restaurants by location, price range, cuisine, and rating.

5. **Restaurant Comparison:** The system should allow users to compare restaurant details.
6. **Review and Rating:** The system may allow users to submit ratings and reviews.
7. **User Management:** The system may allow users to register and log in.
8. **Admin Management:** The administrator should be able to add, update, or delete restaurant information.
9. **Search Results:** The system should display relevant restaurants based on the user's search criteria.

2.2 Non-Functional Requirements

Non-functional requirements describe **how well the system should work**.

1. **Performance:** The system should display search results within a reasonable time.
2. **Usability:** The system should have a simple and user-friendly interface.
3. **Reliability:** The system should provide consistent results when valid data is available.
4. **Security:** The system should protect user information and restrict unauthorized access.
5. **Scalability:** The system should support an increasing number of users and restaurants.
6. **Maintainability:** The system should be easy to update and maintain.
7. **Compatibility:** The system should work on commonly used web browsers and devices.
8. **Accuracy:** The system should display correct restaurant details and calculate rankings according to the defined rules.

2.3 Business Rules and System Constraints

Business Rules

Business rules define the conditions that the system must follow.

1. Restaurants should be ranked according to the selected ranking criteria.
2. Only restaurants matching the user's search criteria should appear in the filtered results.
3. Restaurants with higher scores should generally appear above restaurants with lower scores.
4. Restaurant ratings and reviews should be stored and displayed correctly.
5. Only authorized administrators should be allowed to modify restaurant information.
6. Restaurants with missing information should not be ranked using unavailable values as if they were valid ratings.

System Constraints

System constraints define the limitations of the system.

1. The system requires an internet connection if it is hosted online.
2. Search results depend on the availability and accuracy of restaurant data.
3. The system's performance depends on the server, database, and network connection.
4. The accuracy of restaurant rankings depends on the ranking algorithm and available information.
5. Real-time restaurant information requires a suitable data source or external API.

3. Hardware and Software Requirements

3.1 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB
Storage	10 GB of available space
Display	Standard monitor or laptop display
Internet	Internet connection for online access

These are suggested minimum requirements. Actual requirements may vary depending on the implementation and number of users.

3.2 Software Requirements

The software requirements depend on the technologies selected for the project.

Component	Suggested Software
Operating System	Windows 10/11 or Linux
Frontend	HTML, CSS, JavaScript
Backend	Python with Flask
Database	MySQL
Code Editor	Visual Studio Code
Web Browser	Google Chrome or Microsoft Edge
Testing	Manual testing and browser developer tools

Note: These are suggested technologies. If your project uses a different programming language, framework, or database, replace these items with the technologies you actually use.

4. References

The following resources can be used while developing the Restaurant Searching and Ranking System:

1. Python Documentation – Official Python Documentation
2. Flask Documentation – Official Flask Documentation
3. MySQL Documentation – Official MySQL Documentation
4. HTML Documentation – MDN Web Docs
5. Software Requirements Specification concepts and guidelines.

5. Future Enhancement

The system can be improved in the future by adding the following features:

1. AI-Based Recommendations: Recommend restaurants according to user preferences and search history.
2. Real-Time Location: Find restaurants near the user's current location.
3. Advanced Ranking: Rank restaurants using ratings, reviews, distance, and price.
4. Online Table Booking: Allow users to reserve tables through the system.
5. Map Integration: Display restaurant locations on an interactive map.
6. Mobile Application: Develop an Android or iOS application.
7. Personalized Suggestions: Provide recommendations based on users' previous choices.
8. Live Information: Integrate suitable external APIs to retrieve updated restaurant details when available.

6. Conclusion

The Restaurant Searching and Ranking System helps users find and select suitable restaurants based on their preferences. It provides restaurant search, filtering, comparison, and ranking features in a simple and user-friendly interface.

The system saves users' time and helps them make better decisions when choosing restaurants. In the future, features such as AI-based recommendations, real-time location, and online table booking can make the system more useful and efficient.